MACHINE LANGUAGE.

A Music Production Study Guide

This is a companion to the three-part text adventure trilogy. It's organized to match the games — one section per adventure — followed by an alphabetical glossary for quick lookup while you're working.

Use it three ways:

1. Read it after playing each game to lock in the concepts.
2. Keep it open while you're in Ableton and reach for it when you forget what something means.
3. Hand it to a friend who wants to start making music but doesn't know where to begin.

Music production is a deep field. This guide covers what you need on day one — not what a sound engineer with a master's degree needs. Come back to it as your projects get more ambitious.

GAME 01

MIDI & The Language of Machines

MIDI is the protocol that lets every digital instrument talk to every other digital instrument. Understanding it transforms music software from a confusing mess of menus into a logical system where you know exactly what's happening when you press a key.

What MIDI Is (and Isn't)

MIDI is not sound. MIDI is a stream of **instructions about sound.** When you press a key on a MIDI keyboard, the keyboard doesn't send audio — it sends a message that says *'play note 60 with velocity 100 on channel 1.'* The receiving instrument (a synthesizer, a drum machine, a software plugin) decides what that message actually *sounds* like.

This is why the same MIDI clip can play through a piano sound, a synth bass, or a drum kit and produce three completely different results. The notes are identical. The instrument is different.

The Three Numbers Every MIDI Note Has

NUMBER	RANGE	WHAT IT DOES
Note	0–127	Which pitch (60 = Middle C, 36 = C2, 72 = C5)
Velocity	0–127	How hard the note hits (loud/soft, sometimes tone)
Channel	1–16	Which instrument receives the message

Channels are like radio frequencies. A single MIDI cable can carry 16 conversations at once. Your drum machine listens on channel 1, your bass synth on channel 2, your lead synth on channel 3. Same cable, different conversations.

Drum Maps

Every drum machine and drum sampler has a **drum map** — a chart that says which MIDI note triggers which sound. On most machines, note 36 is the kick, 38 is the snare, 42 is the closed hat, and so on. This is why drawing a note on the lowest row of a drum rack always produces the kick: you're triggering note 36.

The 16-Step Grid

One bar of music in 4/4 time gets divided into 16 steps — each step is a sixteenth note. This is the standard programming grid for drum machines and DAW piano rolls. Every dance beat you've ever heard is built on this grid:

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STEP:  1 . . . 2 . . . 3 . . . 4 . . .
KICK:  ■ . . . . . . . ■ . . . . . . .
SNARE: . . . . ■ . . . . . . . ■ . . .
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HAT:   ■ . ■ . ■ . ■ . ■ . ■ . ■ . ■ .
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Kicks on beats 1 and 3. Snare on 2 and 4 (the 'backbeat'). Hi-hats on every eighth note. Read this pattern: it's the skeleton of techno, EBM, industrial, and most pop music ever written.

Velocity, Quantize, & Humanize

Velocity controls how hard each note hits (0 = silent, 127 = maximum). Programming with all notes at the same velocity makes a beat sound robotic. Mixing in 90s, 110s, and occasional 60s makes it feel alive.

Quantize snaps notes to the grid — useful when you've recorded sloppy MIDI and want it tight.

Humanize is the opposite: it deliberately moves notes slightly off the grid and varies their velocities to mimic the imperfections of a human player. Some music wants this. Some music — industrial, digital hardcore, Atari Teenage Riot — explicitly does *not* want it. The cold rigidity of a perfectly quantized beat is sometimes the whole aesthetic.

GAME 02

Music Fundamentals

Music isn't just sound. It's a body moving through time. Understanding rhythm, tension, and structure matters more than knowing music theory in the academic sense — especially for electronic music, where the theory is often dead simple but the rhythm and arrangement are everything.

Tempo (BPM)

BPM means **beats per minute**. It's the speed of the music — and a physical decision about what kind of motion you want to provoke in the listener.

BPM RANGE	GENRE / FEEL
70–90	Slow / hip-hop / trap
90–110	Downtempo / classic R&B
120–128	House / EBM / sway tempo
130–140	Techno / pump tempo
140–160	Hard techno / trance
160–180	Drum & bass / hardcore
180+	Digital hardcore / gabber

Note Subdivisions

Every beat gets divided into smaller pieces. The faster the subdivision, the busier the rhythm feels.

NOTE VALUE	HITS PER BEAT	FEEL
Quarter note	1	Walking
Eighth note	2	Running
Sixteenth note	4	Sprinting
Thirty-second note	8	Frantic / drum roll

Rule of thumb: the fastest element in your drum pattern controls the perceived speed of the track — even more than the BPM does.

Four-on-the-Floor & the Backbeat

Four-on-the-floor: the kick drum hits on every beat (1, 2, 3, 4). This is the engine of house, techno, EBM, disco, and trance. It works because a kick on every beat lines up with your heartbeat and footsteps — your body literally cannot ignore it.

Backbeat: the snare (or clap) hits on beats 2 and 4. Combined with four-on-the-floor kicks, this is the basic groove of nearly all popular music.

Keys & Scales

A **key** is a family of seven notes that sound good together. A **scale** is the ordered sequence of those notes.

Major keys = bright, happy, resolved.
Minor keys = dark, sad, menacing.

Dark electronic music (EBM, industrial, techno, darkwave) lives in minor keys almost exclusively. If you want your track to sound like Die Sexual or Atari Teenage Riot, write in minor.

Most DAWs (including Ableton 12) will highlight in-key notes for you — turn on Keys & Scales in the piano roll, set it to E minor (or any minor key), and you literally cannot pick a wrong note as long as you stay in the highlights.

Tension & Release — The Master Concept

Music is a series of expectations being created and then either satisfied or denied. When you build up energy, listeners feel **tension**. When you release that energy, listeners feel **pleasure** — actual neurochemical pleasure.

The bigger the tension, the bigger the release. A 32-bar build into a drop feels good because you've made the listener wait. By the time the kick comes back, their body is starving for it.

> *The trick isn't being intense the whole time. The trick is the contrast.*

Song Structure

Most dance tracks follow a similar architecture. Specific bar counts vary, but the principle (tension and release in cycles) doesn't.

SECTION	WHAT HAPPENS
Intro	Elements come in one by one. Tempo locks in.
Build	Tension rises (riser, snare roll, busier hats).
Drop	Everything explodes back in. Release.
Groove	Full track plays out. Body of the song.
Breakdown	Strip elements away. Let the listener breathe.
Build (2nd)	Rise again — usually bigger this time.
Drop (2nd)	The biggest payoff.
Outro	Elements fade out one by one.

The 8-Bar Phrase

Producers think in 8-bar and 16-bar phrases. Most dance music changes *something* every 8 bars — a new element comes in, an old one drops out, a filter sweeps. At 125 BPM, 8 bars is about 15 seconds — the rough limit of human attention before something needs to happen.

Ableton Live 12

Ableton is a tool. The ideas come first. But Ableton has a few unique workflows that no other DAW does as well — once you understand them, you'll see why so many producers swear by it.

Session View vs. Arrangement View

Ableton has two different ways to look at a song. Press **Tab** to switch between them.

VIEW	WHAT IT IS	BEST FOR
Session	A grid of clips. Columns = tracks, rows = scenes. Trigger clips on the fly.	Brainstorming loops, performing live, jamming on ideas.
Arrangement	A traditional left-to-right timeline. Bar 1 to bar 200.	Building a finished song. Editing. Mixing.

The secret weapon: you can hit the global Record button in Session View, perform your set by triggering clips live, and Ableton records the entire performance into Arrangement View. Then you edit the recorded performance into a finished song. No other DAW does this.

The Browser

The left sidebar. Where every sound, instrument, and effect lives. Live 12 added auto-preview — click any sound and it plays through your speakers immediately. You can audition a hundred kick drums in a minute.

Sections to know:

- **Sounds** — instrument presets, organized by category
- **Drums** — drum kits, drum racks, individual hits
- **Instruments** — synths and samplers (Drift, Meld, Wavetable, etc.)
- **Audio Effects** — EQ, compressor, reverb, distortion
- **MIDI Effects** — arpeggiator, scale, chord, random
- **Samples** — raw audio files
- **Packs** — themed sound libraries you can install

Drum Racks

A **Drum Rack** is a 16-pad grid that holds 16 different drum samples. Each pad is mapped to a MIDI note (note 36 = pad 1, note 37 = pad 2, etc.). One MIDI track triggers all the drums.

Almost always preferable to loading single samples on separate tracks. Faster to program, easier to edit, simpler to mix.

The Piano Roll

Where you draw and edit MIDI notes. Time goes left to right. Pitch goes up to down (high pitches at the top). Press **B** to activate the pencil tool. Click anywhere to draw a note.

Below the note grid is a velocity panel — small vertical bars, one per note. Drag them to change how hard each note hits. This is where you humanize a beat.

Audio Effects (Live 12)

Drag effects from the browser onto a track. They process the sound on its way to the master output. Effects work in chains — each feeds into the next. The order matters.

EFFECT	WHAT IT DOES
EQ Eight	Eight-band equalizer. Cut frequencies you don't want, boost ones you do.
Saturator	Adds harmonic distortion. Warmth at low settings, grit at high.
Roar	(New in Live 12) Three-stage saturation/distortion device. Very versatile.
Compressor	Evens out volume. Makes quiet parts louder, loud parts quieter.
Limiter	A hard ceiling on volume. Goes on the master track.
Reverb	Adds space and atmosphere — small room, big hall, infinite cloud.
Delay	Echoes. Useful for everything from subtle width to dub-style chaos.

Automation

Automation is how a parameter changes over time. A filter slowly opening during a build. A volume fading in. A reverb getting louder during a breakdown.

In Arrangement View, click the small triangle on a track header to expand the automation lane. Pick a parameter from the dropdown. Click in the lane to add breakpoints. Drag them to draw a curve.

Day-One Mixing Moves

Mixing is a deep rabbit hole. These four moves alone get you 80% of the way to a track that sounds 'real.'

1. **Volume balance.** Drag the faders so kick and bass are loudest, drums sit just under, melody/pads are softer behind.
2. **EQ — cut, don't boost.** On every track, find the muddy frequency (200–500 Hz usually) and cut it slightly. The mix instantly clears up.
3. **Compress the bass.** Add a Compressor with ratio ~4:1, threshold so the meter shows 3–6 dB of gain reduction. Bass becomes even and powerful.
4. **Limiter on the master.** Drop a Limiter on the master track with ceiling at -1.0 dB. Brings up overall loudness without clipping.

Exporting

File → **Export Audio/Video** (Cmd+Shift+R / Ctrl+Shift+R). Set the loop brace in Arrangement View to cover bar 1 through your final outro bar before exporting.

WAV — uncompressed, biggest, highest quality. Use for sending to labels, DJs, mastering. 24-bit, 44.1 kHz is standard.
MP3 — compressed, small, good enough for sharing online or on your phone. 320 kbps if you have the option.
FLAC — compressed but lossless. Smaller than WAV, identical quality.

ALPHABETICAL

Glossary

Quick lookup for any term across all three games. When you forget what something means in the middle of a session, flip here.

Arpeggio (Arp)

A chord played one note at a time, in sequence. Often hypnotic and repeating. Ableton's Arpeggiator MIDI effect takes a held chord and spits out the notes automatically.

Arrangement View

Ableton's traditional timeline view. Left-to-right bars, horizontal track lanes. Where songs get built and finished.

Audio Effect

A device that processes sound — EQ, compression, reverb, distortion, etc. Lives on a track and modifies whatever passes through it on the way to the master output.

Automation

A parameter that changes over time. Drawn as a curve in the automation lane. Used for filter sweeps, volume fades, sending elements forward and back in the mix.

Backbeat

The snare or clap on beats 2 and 4. Combined with kicks on 1 and 3 (or four-on-the-floor), this is the basic groove of nearly all popular Western music.

Bar

One measure of music. In 4/4 time, a bar contains four beats. The fundamental unit of song length — producers think in 4-bar, 8-bar, and 16-bar chunks.

BPM (Beats Per Minute)

How fast the music goes. 60 BPM = one beat per second. House sits around 120–128, techno 130–140, drum & bass 160–180.

Breakdown

A section of a song where most elements drop out, creating space and tension. Usually followed by a build and a drop.

Browser (Ableton)

The left sidebar where all sounds, instruments, and effects live. Live 12 added auto-preview when you click any sound.

Build (Build-up)

A section where tension rises — risers, snare rolls, busier hi-hats, filter sweeps opening — leading into a drop.

Channel (MIDI)

One of 16 separate 'frequencies' a MIDI cable can carry. Lets one cable control multiple instruments simultaneously.

Chord

Three or more notes played at the same time. In dance music, often broken into arpeggios.

Clip (Ableton)

A single piece of MIDI or audio in Ableton. Lives in a Session cell or on the Arrangement timeline. Can be looped, edited, transposed.

Compressor

An effect that reduces dynamic range — makes loud parts quieter, quiet parts louder. Used on basses, vocals, and sometimes the master to glue a mix together.

DAW (Digital Audio Workstation)

Software for making music. Ableton, Logic, FL Studio, Pro Tools, Reaper. The thing you record, program, and mix inside.

Delay

Echo. Repeats the sound after a set time. Can be subtle (ping-pong) or chaotic (dub delays).

Drift

A new analog-style synth in Ableton Live 12. Available in all editions including Intro. Great for leads, basses, and pads.

Drop

The moment after a build when all the elements come crashing back in. The release of tension.

Drum Map

The chart that says which MIDI note triggers which drum sound. Note 36 is almost always the kick. Different drum machines have different maps.

Drum Rack

An Ableton device that holds 16 drum samples in a grid. Each pad mapped to a MIDI note. One MIDI track triggers all the drums.

EBM (Electronic Body Music)

A genre born in Belgium in the 1980s. Pulsing 4/4 beats, distorted basslines, robotic vocals. Lives at 125–130 BPM. Front 242, Nitzer Ebb, Die Sexual.

Effect Chain

Multiple effects in sequence on one track. Each feeds into the next. Order matters: EQ → Saturator → Compressor → Reverb is different from Reverb → Compressor → Saturator → EQ.

EQ (Equalizer)

An effect that boosts or cuts specific frequency ranges. EQ Eight is Ableton's primary EQ — eight bands you can move and shape independently.

Four-on-the-Floor

The kick drum hitting on every beat. The engine of disco, house, techno, EBM, trance. Locks the listener's body to the tempo.

Frequency (Hz)

How fast a sound wave vibrates. Low frequencies (20–250 Hz) = bass, felt in the body. Mid frequencies (250–4000 Hz) = vocals, melody. High frequencies (4000+ Hz) = air, sparkle.

Groove

The feel of a beat. Comes from velocity variation and slight timing shifts off the rigid grid. What separates a beat that makes you nod from one that doesn't.

Groove Pool (Ableton)

A panel in Ableton where you store and apply groove templates to MIDI clips. One click can transform the feel of a pattern.

Hi-Hat

A small, fast cymbal. Provides the motion in a drum pattern. Usually programmed on eighth or sixteenth notes.

Humanize

Deliberately moving notes slightly off the grid and varying their velocities to mimic human imperfection. Sometimes you want this. Sometimes (digital hardcore) you absolutely don't.

Key

A family of seven notes that sound good together. Major keys = bright. Minor keys = dark. Dark electronic music is almost always in a minor key.

Keys & Scales (Live 12)

A new feature in Ableton 12 that highlights in-key notes in the piano roll. Pick a key (E minor, etc.) and you can't pick a wrong note as long as you stay in the highlights.

Kick Drum

The deepest, most important drum. The gravity of dance music. Sits in the low frequencies (40–80 Hz).

Limiter

A specialized compressor with a hard ceiling on volume. Goes on the master track. Brings up overall loudness without letting anything clip.

Loop

A short pattern that repeats. The basic unit of dance music construction — most tracks are built by arranging and varying loops over time.

Mastering

The final polish on a finished mix. Loudness, EQ balance, stereo width across an entire track or album. Often done by a specialized mastering engineer.

Master Track

The final track in a mix that all other tracks feed into. Where the limiter goes.

Meld

A bi-timbral, MPE-capable synth in Ableton Live 12. Good for textural and experimental sounds.

MIDI (Musical Instrument Digital Interface)

A protocol invented in 1983 that lets digital instruments talk to each other. MIDI sends instructions (play this note, this hard, on this channel) — not sound.

MIDI Effect

A device that processes MIDI before it reaches an instrument. Arpeggiator, Scale, Chord, Random — they manipulate notes, not sound.

Minor Key

A key built on the minor scale. Sounds dark, sad, mysterious, menacing. The default for most EBM, industrial, techno, and darkwave.

Note (MIDI)

A pitch in the MIDI system, numbered 0–127. Middle C is 60. Each integer is one semitone. C2 (note 36) is the kick on most drum machines.

Octave

A doubling of frequency. Two notes one octave apart sound like the 'same note' high and low. MIDI octaves: C0 = 12, C1 = 24, C2 = 36, C3 = 48, C4 = 60 (Middle C), C5 = 72, C6 = 84.

Phrase

A musical unit of 4, 8, or 16 bars. Producers think in 8-bar phrases — change something every 8 bars to keep attention.

Piano Roll

The grid where you draw and edit MIDI notes. Time left-to-right, pitch up-to-down.

Quantize

Snap notes to the grid. Useful for cleaning up sloppy MIDI recording. Can be applied at different resolutions (16th notes, 8th notes, etc.).

Reverb

An effect that simulates the sound of a space — small room, large hall, cathedral. Adds atmosphere and depth.

Riser

A sound that builds in pitch or volume over time, used in build sections to create tension before a drop.

Roar

A new distortion/coloration device in Ableton Live 12. Three saturation stages. Excellent for grit on basses and drums.

Saturator
An Ableton effect that adds harmonic distortion. Subtle warmth at low settings, full destruction at high. Live 12 added a Bass Shaper curve.

Scale
An ordered sequence of notes that defines a key. The minor scale is your friend for dark music.

Scene (Session View)
A row in Ableton's Session View. Triggering a scene fires all the clips in that row simultaneously. Useful for performing different sections of a song.

Semitone
The smallest interval between two notes in Western music. Each MIDI note number is one semitone apart.

Session View
Ableton's grid-based clip launcher view. Where ideas are brainstormed and live performances happen.

Sidechain Compression
A compression technique where one sound triggers compression on another. Most commonly: kick drum triggers volume duck on the bass, so the kick has space to hit. Creates the 'pumping' feel of EDM.

Simpler
An Ableton instrument that loads any audio sample and lets you play it back as a tuned, playable instrument.

Snare
The drum hit on beats 2 and 4 (the backbeat). Slaps your face while the kick punches the floor.

Sub (Sub-bass)
Very low bass frequencies (20–60 Hz). Felt more than heard. The chest-rumble in a club.

Swing
A rhythmic feel where every other note is delayed slightly, creating a loose, strutting groove. Different from rigid quantization.

Tempo
See BPM.

Tension & Release
The master concept of all music. Creating expectations and then satisfying or denying them. Builds create tension; drops release it.

Track (Ableton)
A single instrument or audio source in a session. Has its own clips, effects, and volume fader.

Velocity

How hard a MIDI note hits, on a scale of 0–127. Controls volume and sometimes tone. Varying velocity is essential for groove.

WAV

An uncompressed audio file format. Largest file size, highest quality. The standard for professional delivery to labels and mastering engineers.

Warp (Ableton)

Ableton's time-stretching feature. Lets you fit any audio sample to your project tempo without changing its pitch. The reason Ableton became famous in the DJ world.

ONE LAST THING

You don't learn music production by studying. You learn it by making bad tracks and then making slightly less bad tracks. This guide is here to help you when you forget what something means — not to be memorized.

Open the DAW. Make 30 sketches in your first month. Finish nothing. Steal liberally from the artists you love. You will be surprised, in six months, by what you can do.

> END OF STUDY GUIDE
> GO MAKE NOISE.